



Towards Multimodal User State-Aware Adaptive Guidance in MR

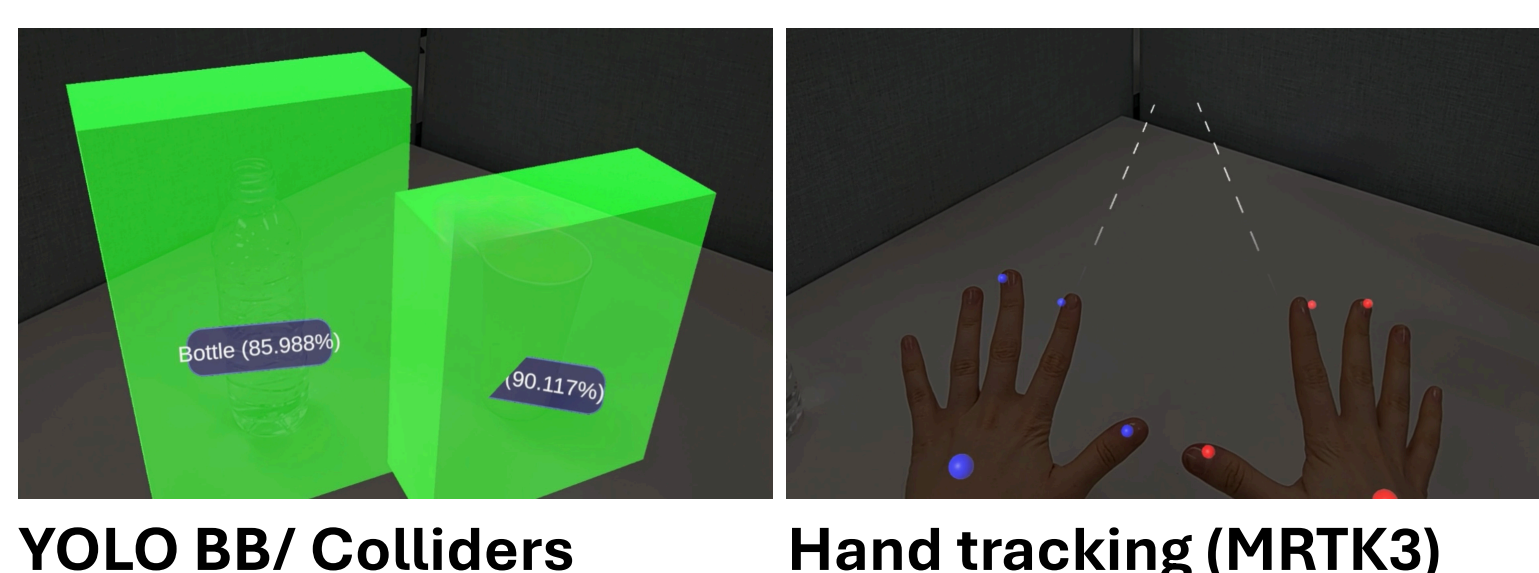
MOTIVATION

Although LLM-based AR assistance shows promise [1], current systems still lack structured task modeling and multimodal user-state tracking required for truly adaptive MR guidance.

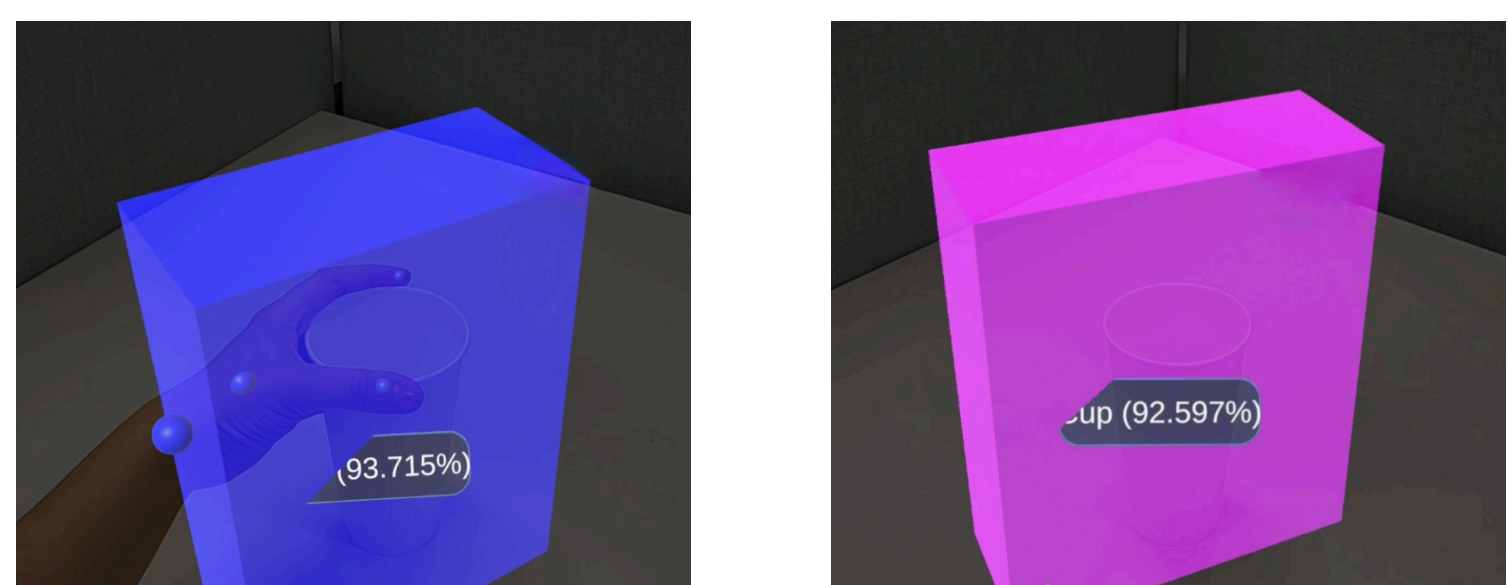
? Can MR guidance systems move beyond static instructions and consider dynamic user behavior and cognitive variability to provide flexible adaptive guidance?

STEP TRACKING IMPLEMENTATION

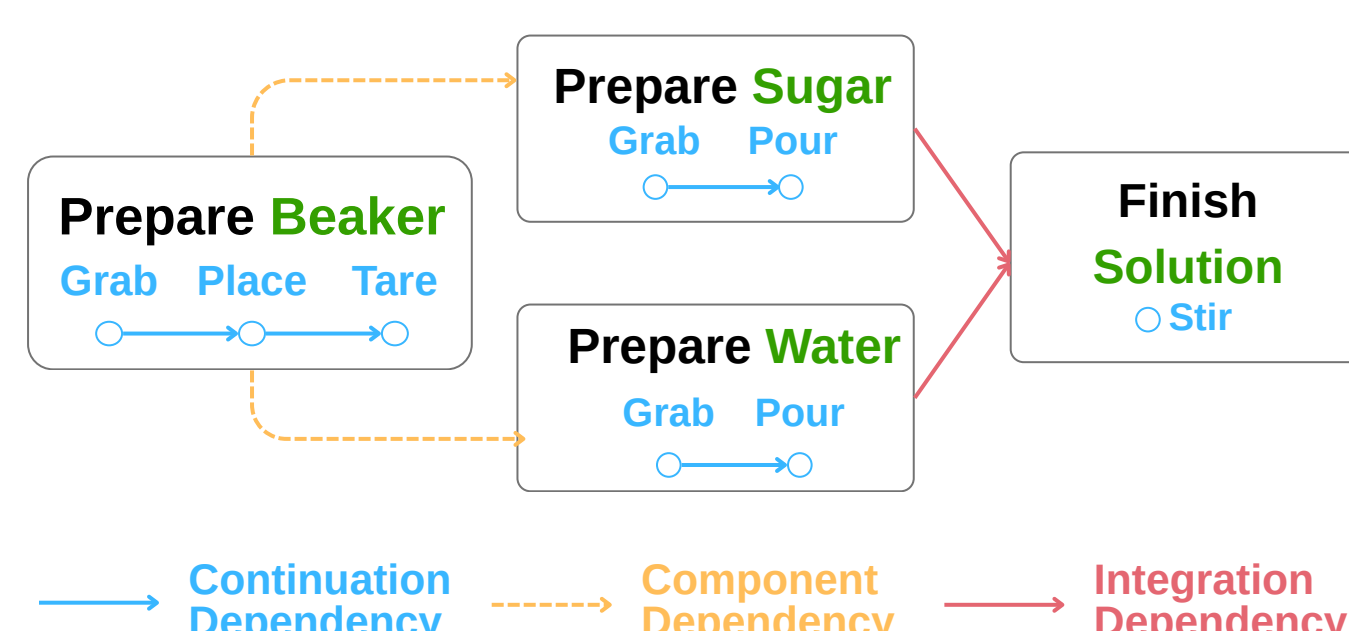
Step 1 - Perception. Yolo-based object detection with MRTK3 kinematics tracking (e.g., hands-related grab, release, poke, tilt and gaze-related enter, exit, dwell time).



Step 2 - Action Primitive Inference. Heuristic temporal rules over kinematic actions (e.g., hand joints enter collider + grab gesture + move + release gesture = place).



Hands-Object Interaction Gaze-Object Interaction



LLM-Generated Task Graph (DAG) example

Step 3 - Step Validation. Recognized actions are validated against the task **DAG** to determine correctness and progress (e.g., expected, incorrect, premature actions).

```
grab: Cup | tilt=6.9°
release: Cup | tilt=17.5° (near: Bottle)
X POUR: Cup → Bottle
  incorrect_unrecoverable: X STOP! This action is dangerous.
  Progress: 0/2

grab: Bottle | tilt=32.6°
release: Bottle | tilt=49.9° (near: Cup)
O POUR: Bottle → Cup
  premature: O Not yet! First, place bottle near cup.
  Progress: 0/2

grab: Bottle | tilt=89.2°
release: Bottle | tilt=88.0° (near: Cup)
O PLACE: Bottle → Cup
  expected: ✓ Place Bottle near Cup. Next: pour from bottle into cup.
  Progress: 1/2
```

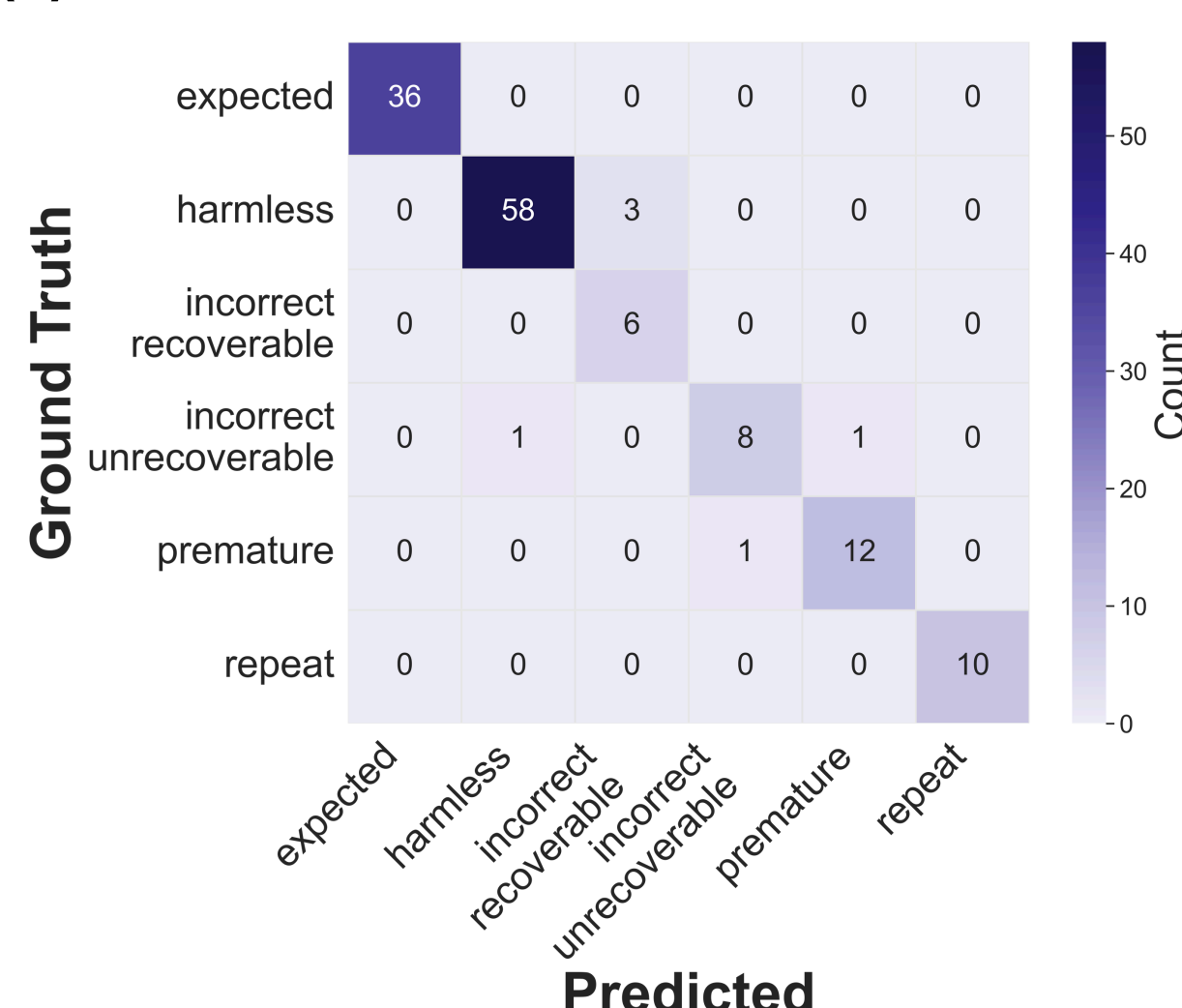
Outcome Classification Examples

PRELIMINARY RESULTS

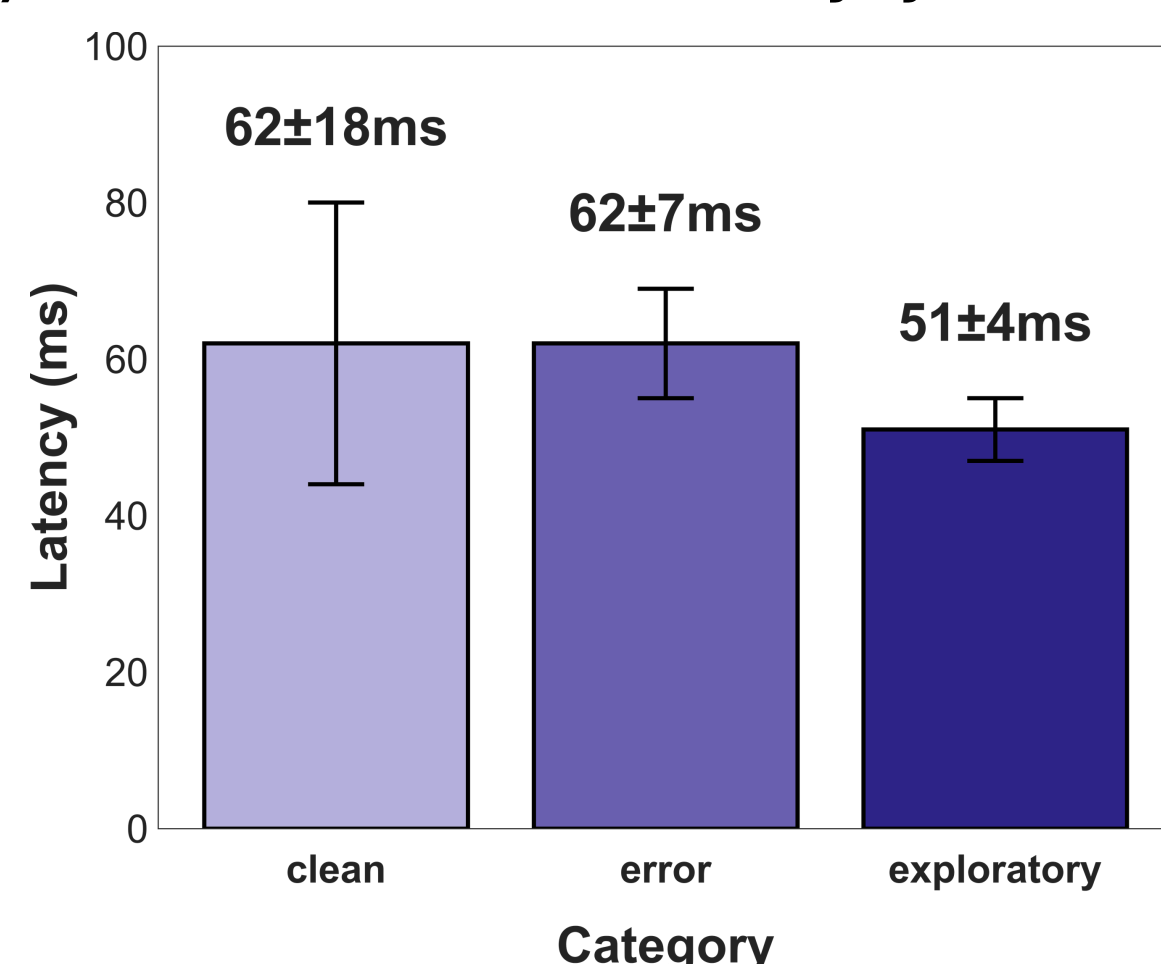
Real-Time step classification performance was tested across **Clean** (intended protocol-following), **Error** (including predefined mistakes) and **Exploratory** (free-form exploratory behavior) sessions.

Category	Action Recognition, F1	Outcome Classification, F1	Deviation Detection Latency (ms)
Clean (n=2)	1.00	1.00	116 ± 89
Error (n=2)	0.94	0.93	69 ± 45
Exploratory (n=2)	0.97	0.96	45 ± 29
Mean (n=6)	0.97	0.96	77 ± 65

(a) Outcome classification



(b) Action-to-classification latency by session type



CONCLUSION AND FUTURE WORK

- LLM-generated task graphs combined with semantic step tracking enable robust, low-latency task monitoring in MR (F1 = 0.97, ≤ 62 ms), including off-protocol behavior.
- Future work will extend this small-scale evaluation (n = 6) by integrating temporal user-state modeling and closed-loop adaptive guidance.

REFERENCES

- [1] M. Qorbani and K. Paynabar. Teaching llms to see and guide: Context-aware real-time assistance in augmented reality, 2025.
- [2] J. J. Yang, L. Qiu, E. A. Corona-Moreno, and L. Shi. Amma: Adaptive multimodal assistants through automated state tracking and user model-directed guidance planning, 2024.
- [3] Y. Hou, Q. Xie, N. Zhang, and L. Jian. Cognitive load classification of mixed reality human computer interaction tasks based on multimodal sensor signals. Scientific Reports, 2025.
- [4] M. Pluisch, J. Gugenheimer, Y. Cho, S. Julier, and E. Kruijff. A multisensor approach for cognitive load assessment in mobile augmented reality, 2025.

SYSTEM OVERVIEW

